

HUNTER

AUTONOMOUS TARGET INTERCEPTION

Track: **Mobile Robotics & Autonomous Systems**



THREE PROGRESSIVELY DIFFICULT MISSIONS

OVERVIEW

Autonomous robots often operate in environments where the position of a target is uncertain and its future motion is unknown. A robot may need to intercept a moving vehicle, pursue another autonomous system, or navigate through a cluttered environment using incomplete information. In such situations, simply moving toward the target's current position is not sufficient. The robot must interpret the available information, estimate the target's motion, and choose an effective interception strategy.

In **HUNTER**, participants control an autonomous ground robot inside a simulated 3D arena. A second robot, called the **Runner**, acts as the target. The Hunter must navigate the environment and reach the Runner using only the information available through the simulation. The challenge consists of three progressively difficult missions: **FIND** → **PREDICT** → **HUNT**.

CHALLENGE

Your task is to develop an autonomous controller for the Hunter robot.

The Hunter operates on a planar surface inside a visually simulated 3D environment. The robot can control its forward motion and turning subject to the limits defined by the simulation. The Runner is controlled by the environment; its exact future motion is not provided to the participant.

Depending on the mission, the Runner may be stationary or continuously moving. When the Runner is moving, the Hunter does not receive its exact velocity or future trajectory. Instead, the Hunter receives position observations at specified intervals. These observations may contain noise and delay. The Hunter must use the available information to determine how to move and ultimately intercept the Runner.

ARENA & ROBOT MODEL

The simulation consists of a bounded arena containing the **Hunter robot**, the **Runner robot**, **static obstacles** where applicable, and **arena boundaries**. Although the environment is rendered in 3D, the robotics problem is planar. The Hunter's state is represented by

$$(x, y, \theta)$$

where x and y represent position and θ represents heading. The Hunter accepts commands corresponding to **linear velocity** and **angular velocity**, and the simulation enforces the defined limits on both. The environment is responsible for robot motion, collision handling, target motion, sensor generation, and evaluation.

MISSIONS

MISSION 01 — FIND

Navigate to a Stationary Target

The Runner remains stationary. The arena contains static obstacles that prevent the Hunter from taking a direct path to the target. The Hunter must navigate from its starting location to the Runner while remaining inside the arena and avoiding obstacles.

OBJECTIVE

Determine and execute an efficient collision-free route from the Hunter to the stationary Runner. Participants are free to choose their own planning and control approach.

MISSION 02 — PREDICT

Intercept a Moving Target

The obstacles are removed. The Runner now moves continuously through the arena. The Hunter does not receive the Runner's actual velocity, acceleration, future trajectory, or future position. Instead, it receives intermittent observations of the Runner's position. The observations may be **noisy**, **delayed**, or separated by a **sampling interval**. The Hunter must infer the motion of the Runner from the available observations and use that information to intercept it.

OBJECTIVE

Estimate the Runner's motion and determine a suitable future interception location. A solution that simply follows the most recent observation may not be sufficient. The Hunter must account for the fact that the Runner continues moving while the Hunter is approaching it.

MISSION 03 — HUNT

Intercept a Moving Target Through Obstacles

The final mission combines the previous challenges. The Runner is moving continuously while static obstacles are present in the arena. The Hunter receives intermittent and imperfect observations of the Runner while simultaneously having to navigate around obstacles. The Hunter must continuously make decisions while respecting its motion limits and avoiding obstacles.

OBJECTIVE

Successfully intercept the moving Runner while maintaining safe and efficient navigation through the environment. The Hunter successfully completes a mission when it reaches the Runner within the capture region defined by the simulation.

CONTROLLER

Participants will be provided with a controller interface through which the simulation communicates with their solution. The controller receives the information available to the Hunter and returns the Hunter's motion command. Participants are required to develop three controllers:

- `controller_mission1.py` — For FIND
- `controller_mission2.py` — For PREDICT
- `controller_mission3.py` — For HUNT

Each controller may use its own internal logic and mathematical approach.

EVALUATION

Each mission is evaluated independently using the official simulation environment. Performance is evaluated using measurable criteria including:

Mission 1 — FIND

Successful navigation · Collision-free operation · Route efficiency

Mission 2 — PREDICT

Successful interception · Interception time · Target estimation and prediction performance

Mission 3 — HUNT

Successful interception · Collision-free navigation · Interception time · Overall interception efficiency

The final evaluation may use scenarios that are different from the public development scenarios.

HIDDEN EVALUATION

Public scenarios are provided for development and testing. The final evaluation may use different starting positions, obstacle layouts, target trajectories, observation conditions, and environmental configurations. The underlying rules and constraints remain unchanged.

Participants should therefore design controllers that generalize to new scenarios rather than relying on predefined paths or hard-coded solutions.

DELIVERABLES

Each team must submit:

1. **Mission 1 Controller** — `controller_mission1.py`
2. **Mission 2 Controller** — `controller_mission2.py`
3. **Mission 3 Controller** — `controller_mission3.py`
4. **README** — a brief description of: solution approach, assumptions, how to run the controllers.
5. **Technical Report** — the report should describe: mathematical model, reasoning and approach, estimation/prediction logic, navigation strategy, important design decisions.

Teams should be able to explain their submitted solution during evaluation.

FINAL OBJECTIVE

The challenge is to develop a Hunter capable of progressing from simple navigation to autonomous interception of a moving target. A successful solution combines autonomous decision-making, mathematical reasoning, and effective robot control to intercept a target whose future motion is not directly known.

FIND → PREDICT → HUNT